



Acne Vulgaris

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Acne Vulgaris:

Learning Outcomes:

The pathogenesis of acne.

The clinical types of acne .

Assessment of the severity of acne.

The treatments of different types of acne.



Acne Vulgaris

Acne vulgaris is a multifactorial disorder of the pilosebaceous unit.

The clinical picture can vary significantly, from mild comedonal acne to fulminant systemic disease.

acne affects ~85% of young people between 12 and 24 years of age and is therefore mostly it is a physiologic occurrence in this group.

With the general trend over the past few decades for an earlier onset of puberty, preadolescent acne affecting children 7 to 11 years of age has also become more common.

can have acne with this arms
in sebaceous area
this is called non-inflammatory
could be white (close), black (open) comedones



Genetic Factors:

It is widely believed that the tendency to have substantial acne (including the nodulocystic variant) runs in families, and an association between moderate-to-severe acne and a family history of acne has been observed in several studies.

People who have severe acne often have parents with severe acne history



Dietary Factors:

The relationship between diet and acne remains controversial.

The intake of milk, especially skim milk, is positively associated with acne prevalence and severity. *Usually not cause it but precipitates it*

Whey protein supplements for bodybuilding has also been reported with Vitamin B12 supplementation can potentially trigger the development of acne. *junk food, fast food*



Pathogenesis of acne:

The development of acne involves the complex interplay of four main factors:

not necessarily all occur/occur together

- (1) follicular hyperkeratinization.
- (2) hormonal influences on sebum production and composition.
- (3) inflammation.
- (4) *C. acnes* (cutibacterium acnes) residing in the sebaceous follicle.



Clinical Features:

Acne is typically found in sites with well-developed sebaceous glands, most often the face and upper trunk.

acne lesions are divided into non-inflammatory and inflammatory groups based upon their clinical appearance.

Non-inflammatory acne is characterized by open and closed comedones.

Inflammatory acne is characterized by papules, pustules, and nodules of varying severity.

mainly comedones



papules & pustules





Acne Variants:

Post-adolescent acne in women:

Inflammatory acne beyond 25 years of age is most common in women and may be associated with a high level of psychological stress. *usually one of two lesions appear related to menstrual cycle → exacerbated with psychological stress and worsened*

The mandibular area is involved in ~80% of women with acne.

Premenstrual flares are common.

Up to 30% of those with (irregular menses) *→ could be PCOS* have other signs of hyperandrogenism, such as hirsutism and androgenetic alopecia.



Acne Variants:

Acne fulminans:

Acne fulminans is the most severe form of acne and is characterized by the abrupt development of nodular and suppurative acne lesions in association to systemic manifestations.

This uncommon variant primarily affects boys 13–16 years of age.

Systemic manifestations include fever, arthralgias, myalgias, hepatosplenomegaly, and severe malaise.

→ so be careful if there's acne with skin manifestations



Acne Fulminans:

Laboratory abnormalities vary and include an elevated ESR, proteinuria, leukocytosis, and anemia.

Recommended treatment of acne fulminans includes prednisone as monotherapy for at least 2–4 weeks, followed by initiation of low-dose isotretinoin after the acute inflammation subsides. *↳ for treating acne*

↳ systemic inflammation (not for treating the acne/it actually cause acne)



Acne Variants:

Acne conglobata and associated conditions:

Acne conglobata is a severe form of nodulocystic acne that may have an eruptive onset but without systemic manifestations.

Acne Variants:

Solid facial edema: *due to repetitive poorly treated facial inflammation*

An unusual and disfiguring complication of acne vulgaris is solid facial edema (Morbihan disease).

Clinically, there is a distortion of the midline face and cheeks due to soft tissue swelling.

Impaired lymphatic drainage and fibrosis.

Treatment with isotretinoin for 4–6 months has been reported to lead to improvement.

Acne Variants:



Acne fulminans *if there's systemic symptoms*

no systemic system → acne conglobata



Acne Variants:

Neonatal cephalic pustulosis (neonatal acne):

Neonatal cephalic pustulosis occurs in more than 20% of healthy newborns. Lesions usually appear at about 2 weeks of age and generally resolve within the first 3 months of life.

Because of transient and benign nature of this eruption, parental reassurance alone is usually adequate.

→ DO NOTHING

maternal estrogen



Acne Variants:

Infantile Acne: *→ they should be treated promptly to high risk of scar*

Infantile acne typically has its onset at 2–12 months of age. In contrast to neonatal cephalic pustulosis, comedon formation is prominent and pitted scarring may develop in up to half of patients.

Its pathogenesis reflects the androgen *↑ androgen production* production intrinsic to this stage of development.

usually resolves within 6–18 months and remains quiescent until around puberty, with an increased risk of severe acne during adolescence. Topical retinoids and benzoyl peroxide are first-line treatments for infantile acne.



Acne Variants:

Mid childhood acne:

Acne presenting between 1 and 6 or 7 years of age.

There is possibility of an underlying hyperandrogenic condition such as premature adrenarche, congenital adrenal hyperplasia, or an androgen-secreting tumor should be considered.

signs of hyperandrogenism and precocious puberty, and the child's growth curve should be carefully reviewed.



Acne Variants:

Preadolescent acne:

It is common for acne to begin to develop with the onset of adrenarche, typically between 7 or 8 to 11 years of age, which is often prior to other signs of pubertal maturation

Preadolescent acne tends to be primarily comedonal and favors the forehead and central face (“T-zone”).

Treatment is similar to that for adolescent acne.



Acne Variants:

Acne excoriée: *→ kinda type of post-adolescent acne*

Acne excoriée occurs more often in young women. Typical comedones and inflammatory papules are systematically excoriated in a ritualistic manner, leaving crusted erosions that may scar.

Linear erosions suggest self-manipulation, and an underlying psychiatric component should be considered.

**excessive use of makeup/treatment can cause it*



Acne Variants:

Acne associated with endocrinologic abnormalities:

hyperandrogenism should be suspected in women and late adolescent girls with irregular menstrual periods.

Polycystic ovary syndrome (PCOS) is the most common condition associated with an elevated serum testosterone. *→ clinically lower part of face, symmetrical menstrual cycle, hirsutism, androgenic alopecia*

Treatment of this condition is according to the etiology.

Pathology:

early lesions, microcomedones are seen. A mildly distended follicle with a narrowed follicular opening is impacted by shed keratinocytes.

In closed ^{white color} comedones, the degree of follicular distension is increased and a compact cystic structure forms.

Within the cystic space, eosinophilic keratinaceous debris, hair, and numerous bacteria are present.

Open comedones ^{black color due to oxidation} have broad, expanded follicular ostia and greater overall follicular distension.



Acne:

Differential Diagnosis:

Milia. *✓ keratin accumulated cyst*

Sebaceous hyperplasia.

Syringoma.

Trichoepithelioma.

Rosacea.

Perioral dermatitis.

Folliculitis.

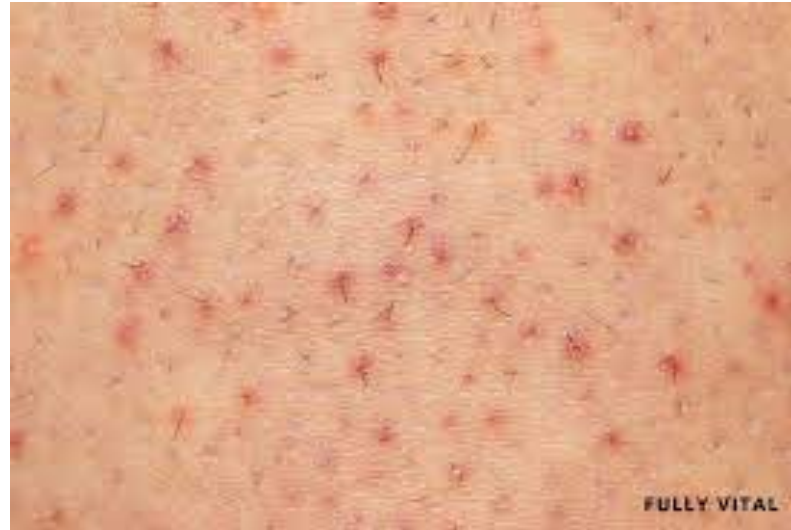
Miliaria rubra.

↳ mainly in children

Acne:



Acne



folliculitis

Treatment:

According to the severity of acne, the treatment differs,

Mild: Several papules, pustules, and/or comedones; no nodules.

Moderate: Multiple papules and pustules; few scattered nodules; variable comedones.

Severe: Numerous papules and pustules; multiple nodules; variable comedones.

→ usually at least one cyst

Very severe: Numerous nodules with conglobate or hemorrhagic lesions; $\pm$ systemic symptoms (acne fulminans).

Severe



mild





Treatment:

Topical Treatments:

Topical retinoids.

Topical antimicrobials like: benzoyl peroxide, azelaic acid.

topical antibiotics: (clindamycin, erythromycin, minocycline).

Sodium sulfacetamide, Topical dapsone 5% and 7.5% gels.

Salicylic acid is a widely used comedolytic agent.

Niacinamide (nicotinamide) can decrease sebum and increase ceramide.



Treatment:

Systemic Therapy:

1. Antibiotics:

First-line therapy for moderate to severe inflammatory acne includes use of oral tetracyclines, primarily doxycycline, or minocycline.

↳ unless inadequate or contra-indicated

Doxycycline and minocycline are deemed equally effective in the treatment of the inflammatory component of moderate to severe acne.



Treatment:

Oral macrolide antibiotics, erythromycin and azithromycin, can be used for acne in patients who are unable to tolerate tetracyclines, pregnant, or <8 years of age.

↳ tetracyclines are
contraindicated

↳ affecting tooth
development

Trimethoprim-sulfamethoxazole may also be of benefit for acne, but its use is discouraged due to the greater risk of severe adverse reactions.



Treatment:

2. Hormonal therapy:

Combined oral contraceptive pills, which contain an estrogen plus a progestin, block both ovarian and adrenal production of androgens.

usually second line to contraception
The antiandrogen cyproterone acetate and flutamide are androgen blockers used in treating acne.

Spironolactone functions as an androgen receptor blocker and, in doses of 50–100 mg twice daily, has been shown to reduce sebum production and improve acne.



Treatment:

3. Isotretinoin:

It is ideal for patients with severe, nodulocystic acne refractory to treatment including oral antibiotics.

Nowadays it can be useful in other types of acne, these include significant acne that is resistant to therapy

(including oral antibiotics), relapses quickly, and/or results in scarring, as well as Gram-negative folliculitis, pyoderma faciale, and acne fulminans.



Treatment:

The most common adverse effects of isotretinoin involve the skin and mucous membranes and are dose-dependent. These include cheilitis, dryness of the oral and nasal mucosa, generalized xerosis, or infections. *before treatment send for routine bloodwork*

Teratogenicity is a serious potential complication.

It may increase serum triglyceride and/or cholesterol.

There is an association between isotretinoin treatment and increased risk of depression;

also send them every month

- CBC
- AST
- ALT
- LFT
- Lipid profile
- Pregnancy



Treatment:

4. Surgical Treatment:

Like comedo extraction, and laser therapy for the sites of acne and the scars left.

Sometimes chemical peels may be necessary to treat the sites of acne .



Reference:

Bologna text book 5th edition 2024



Aacne Vulgaris:

Questions?

